

≡ Hide menu

Week 4: Optical Flow

✓

Ungraded Plugin: 4.1

Overview of Optical Flow

4 min

✓

Ungraded Plugin: 4.2

Motion Field and Optical Flow

10 min

✓

Practice Assignment: 4.2

Motion Field and Optical Flow Self-Check Quiz

Submitted

✓

Ungraded Plugin: 4.3

Optical Flow Constraint Equation

16 min

✓

Practice Assignment: 4.3

Optical Flow Constraint Equation Self-Check Quiz

Submitted

✓

Reading: 4.4 Video

Correction

10 min

<>

Ungraded Plugin: 4.4

Lucas-Kanade Method

10 min

📋

Practice Assignment: 4.4

Lucas-Kanade Method Self-Check Quiz

3h

<>

Ungraded Plugin: 4.5

Coarse-to-Fine Flow Estimation

8 min

📋

Practice Assignment: 4.5

Coarse-to-Fine Flow Estimation Self-Check Quiz

3h

<>

Ungraded Plugin: 4.6

Application of Optical Flow

6 min

📋

Practice Assignment: 4.6

Application of Optical Flow Self-Check Quiz

3h

💬

Discussion Prompt: Week

4 Questions and Feedback

Week 4 Quiz

4.4 Video Correction

In [video 4.4 Lucas-Kanade Method](#) [↗](#), from 1:36 to 2:56, the right matrix is updated (see below).

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

Lucas-Kanade Solution

For all points $(k,l) \in W$: $I_x(k,l)u + I_y(k,l)v + I_t(k,l) = 0$

Let the size of window W be $n \times n$

In matrix form:

$$\begin{bmatrix} I_x(1,1) & I_y(1,1) \\ I_x(k,l) & I_y(k,l) \\ \vdots & \vdots \\ I_x(n,n) & I_y(n,n) \end{bmatrix}$$

A
(Known)
 $n^2 \times 2$

$$\begin{bmatrix} u \\ v \end{bmatrix}$$

$\mathbf{u}$
(Unknown)
 2×1

$$= \begin{bmatrix} -I_t(1,1) \\ -I_t(k,l) \\ \vdots \\ -I_t(n,n) \end{bmatrix}$$

B
(Known)
 $n^2 \times 1$

n^2 Equations, 2 Unknowns: Find Least Squares Solution

Go to next item

✓ Completed

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